Process Efficiency as a Barrier to Innovation

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Abstract

I build a dynamic model of firm innovation in which product innovation and process efficiency *jointly* determine whether an innovator can displace an incumbent. Process efficiency represents the organizational capabilities that firms choose to invest in, and since it applies to every product line a firm owns, large firms invest more in it in equilibrium. The model predicts that R&D subsidies to large firms lead to more investment in process efficiency, which in turn blocks product innovation by smaller firms. I find evidence for this prediction in firm-level data from the 2008–2020 Chinese Tax Survey: within industries, and holding small firms' own subsidies fixed, a 10-percentage-point increase in the subsidy share of large firms lowers small-firm patenting by roughly 7%. This finding is robust to shift-share instrumentation using provincial patterns in subsidy allocation. Calibrated to the Chinese economy, where subsidies currently favor large firms, the model implies that removing all subsidies raises consumption-equivalent welfare by 1.0%, while reallocating the existing subsidy budget along the size distribution raises it by 10.7%.

Keywords: innovation subsidies, process efficiency, creative destruction, firm size, misallocation, China
JEL Codes: O31, O38, O41, L25

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1 Introduction

EMI, the British music company, invented the CT scanner, and its engineer Godfrey Hounsfield shared a Nobel Prize for it. However, EMI had little experience in medical equipment, and it struggled to manufacture the scanner at scale and to support the machines once they were in hospitals. This allowed General Electric, which had been selling and servicing X-ray machines for decades, to take over the market within ten years (Teece, 1986). This example illustrates a general point. Having a better product alone does not necessarily mean a firm can win over the market. The innovating firm needs to manufacture and sell it efficiently, and the production, distribution, and organizational routines that make this possible can take years to build. I call these capabilities *process efficiency*.

Process efficiency is absent from standard models of creative destruction. In the quality-ladder models that follow Klette and Kortum (2004), product quality alone determines which firm serves a market. The producer offering the best version of a product can serve the market and is only displaced when another firm develops a higher-quality version. Models that do include process efficiency typically treat it as a fixed firm characteristic (Aghion et al., 2025). However, survey evidence suggests that process efficiency is something firms invest in. Cohen and Klepper (1996) find that firms spend a substantial share of their R&D on process improvement, and that this share rises with firm size. They attribute this to cost spreading. The idea is that a process improvement typically applies to every product the firm produces, so its return increases with the number of product lines. Much of the cost, however, is incurred only once, since the improvements contribute to systems and practices shared across the firm. Managerial knowledge is one example: the knowledge held by managers at the top can be used by every worker down the hierarchy, allowing the cost of knowledge to be spread over the firm's entire output (Garicano, 2000; Caliendo and Rossi-Hansberg, 2012). Software is another example, since it is written once and used across every product line a firm operates (De Ridder, 2024; Lashkari et al., 2024).

This paper develops a model in which product innovation and process efficiency jointly determine a firm's competitiveness on each of the products it sells. A challenger with a better product displaces an incumbent only if the quality improvement is large enough to offset the incumbent's cost advantage. Since process efficiency rises with firm size, challengers need larger quality improvements to displace larger incumbents, and an innovation that would succeed against a small incumbent may fail against a large one. Small firms and entrants face the greatest difficulty because they have little process efficiency of their own. The incumbent earns the operating return on its process investment, but it does not bear the cost of the innovations it prevents, which would otherwise have raised growth.

The model is an otherwise standard quality-ladder growth model (Klette and Kortum, 2004; Akcigit and Kerr, 2018) in which firms divide their R&D between product innovation and process efficiency. A firm accumulates product lines and becomes large through product innovation, and it invests in process efficiency to make its lines hard to displace. There are no ex ante differences in the ability of firms, so size is the only reason why some firms are more process-efficient than others.

In the model, subsidies to large firms finance additional process R&D. The efficiency it buys raises the bar on the lines those firms hold, which in turn crowds out product innovation by smaller firms and entrants. The model therefore predicts that small-firm patenting should fall if subsidies become more concentrated among large firms.

I test this prediction using the Chinese Tax Survey linked to patent applications from the China National Intellectual Property Administration, a panel of roughly 1.9 million firm-year observations on R&D-active firms from 2008 to 2020. Within each 3-digit industry $\times$ year cell, I regress patenting by size group on the share of subsidies received by firms in the top employment decile, controlling for each group's own subsidy intensity and firm count. I find that a 10-percentage-point increase in the top-decile share of subsidies predicts a fall of roughly 7% in small-firm patenting relative to medium-firm patenting in the same industry-year. Patenting by large firms shows no measurable increase.

One potential concern is strategic patenting, since firms may file patents to qualify for tax preferences, subsidies, or procurement eligibility. If that is the case, the decline in small-firm patenting may not reflect a decline in true inventive activity. I find that the results hold when I restrict the outcome to breakthrough patents identified from patent text. Moreover, I control for revenue concentration because industries in which large firms receive most subsidies may also be industries in which they account for most revenue, and that dominance could itself depress small-firm patenting. With this control, the subsidy coefficient is estimated conditional on the market shares of large firms. Finally, I address the endogeneity of subsidy allocation with a Bartik instrument. The instrument makes use of the fact that subsidies in China are administered at the provincial level, and that provinces differ persistently in their tendency to favor large firms. It weighs each province's large-firm subsidy bias by the industry's pre-period employment share in that province. The instrumented estimate for small firms remains negative and significant. Taken together, the evidence is consistent with the model's prediction that concentrating subsidies among large firms crowds out innovation by small firms.

Large firms in the model devote a greater share of their R&D to process improvement. The National Enterprise Innovation Survey, a census of above-scale firms conducted by the National Bureau of Statistics, corroborates this pattern. Among product innovators, the share that also reports process innovation rises in each successive size category, from 65% of micro firms to 87% of large firms. Trade-secret adoption rises over the same size range, from 8% to 37%, so larger firms are also more likely to protect innovation in a form that requires no patent disclosure.

Existing innovation subsidies in China are heavily concentrated among large, established incumbents. Program eligibility rules and application procedures tend to favor firms with scale, as do the provincial governments that administer the funds. Between 2008 and 2020, firms with more than 10,000 employees received 59% of innovation subsidies but produced only 12% of patents. Patent output per subsidy yuan is also lowest among the largest firms (Online Appendix A). Measuring the innovation and welfare lost under the current allocation, and determining how subsidies should be reallocated, would require a structural model. It would capture the general equilibrium effects of reallocating subsidies, including changes in

wages, entry, and the bar facing challengers.

I estimate the model by simulated method of moments on the panel of R&D-active firms from the Chinese Tax Survey. The estimation targets seven moments that describe innovation and firm size in China: aggregate growth, the entry rate, the entrant employment share, the slopes of log(R&D/Revenue) and log(Pat/R&D) on log employment, the entry R&D share, and the variance of log employment. The model fits the targeted moments closely and matches several nontargeted patterns in the data. As an out-of-sample check, I regress small-firm patenting on subsidy concentration in model-simulated data and obtain a coefficient slightly more than half the reduced-form estimate.

To assess whether process efficiency is needed to fit the data, I compare the full model with a restricted version that shuts off this mechanism. The moment that disciplines it is the slope of log(Pat/R&D) on log employment. In the Chinese sample the slope is -0.56, which means that large firms produce fewer patents per yuan of R&D. Without process efficiency, every firm in the model makes the same number of attempts at the same cost, so the ratio of patents to R&D does not vary with firm size, as in Klette and Kortum (2004). The restricted model therefore predicts a flat gradient, and it is the negative gradient in the data that identifies the process-efficiency margin.

I then use the calibrated model to evaluate three counterfactual subsidy schedules. The first sets all subsidy rates to zero and asks whether the existing regime is welfare-improving relative to a baseline with no subsidies at all. In a first-best environment where product and process R&D are separately observable and contractible, a social planner would subsidize or tax product innovation and process efficiency differentially along the size gradient, in order to internalize the positive externality from product innovation and the negative externality from the blocking loss. In reality this policy is difficult to implement, because R&D composition is often not directly observable and because firms can shirk on product-innovation effort or relabel process investments as product R&D. In a second-best environment the social planner could instead contract on firm size, which in the model predicts R&D composition. The second and third counterfactuals therefore optimize over this policy class. In the budget-neutral exercise the planner reallocates the existing subsidy budget across the size distribution; in the variable-budget exercise the planner chooses both its total subsidy spending and its allocation.

The existing regime is welfare-reducing: removing all innovation subsidies raises welfare by 1.0% in consumption-equivalent terms. In the budget-neutral exercise, the welfare-maximizing allocation removes all subsidies from incumbents and concentrates them on entrants, and welfare rises by 10.7%. The gains are even larger in the variable-budget exercise, where the optimal policy subsidizes entrants and the smallest incumbents and leaves larger incumbents unsubsidized. Welfare rises by 18.2% and aggregate growth by 1.2 pp.

These welfare gains are conservative for two reasons. First, they capture the firm-size margin in isolation, in a model where firms are ex ante identical. A planner able to condition on founder ability, industry, or product characteristics may achieve larger gains. Second, the model assumes that process R&D produces no patents. If some process improvements are patented, those patents would come mainly from large firms,

where process R&D is concentrated. The decline in product patents per unit of R&D with firm size would then be steeper than the observed decline in total patents. Because this slope disciplines the mechanism, the estimation would infer too little process R&D and understate how difficult large firms are to displace. The reported gains would therefore understate the benefits of reallocating subsidies.

This paper builds on three literatures. The first is the quality-ladder theory of firm dynamics. Klette and Kortum (2004) provide the canonical framework, and Akcigit and Kerr (2018) extend it by distinguishing internal innovation, which improves a firm's existing lines, from external innovation, which captures new ones. I distinguish instead between product innovation and process efficiency, and I bring the cost-spreading logic of Cohen and Klepper (1996), who document that large firms do relatively more process R&D and explain it in partial equilibrium, into a general-equilibrium framework, where it produces the blocking externality. Aghion et al. (2025) also separate the two activities in general equilibrium, but they treat process efficiency as an exogenous firm trait that raises markups and does not affect creative destruction.

The second is the literature on innovation policy and firm dynamics. In Acemoglu et al. (2018), subsidies to incumbents reduce growth by keeping low-type firms alive. In my model the loss comes from the R&D choices of large firms, which are no less capable than others, so even a subsidy to the most innovative firms can reduce welfare if those firms are large. Meanwhile, my finding goes against the conclusion of Aghion et al. (2025) that size-dependent subsidies have limited value. This is because I allow process efficiency in my model to be a choice that depends on size rather than a fixed trait, and it is the size margin that drives the blocking externality in the model, which is consistent with evidence from Chinese firm-level data. Size-dependent reallocation of subsidies is hence useful in mitigating the externality. Aghion et al. (2023) also let process efficiency raise a firm's quality-adjusted productivity on every line it operates, but in their model the quality step from a successful innovation is always larger than the efficiency gap between firms, so that innovations always succeed and process efficiency affects only markups. Therefore, their model has no blocking mechanism and hence does not rationalize my empirical finding that subsidy concentration at large firms blocks out innovation by smaller firms. Finally, this paper is related to Atkeson and Burstein (2019), who study the aggregate effects of firm-level innovation policy but abstract from the composition of R&D.

The third is the empirical literature on innovation in China, which documents the growth of Chinese patenting (Wei et al., 2017), the role of intellectual-property protection and ownership in shaping it (Fang et al., 2017), and the effects of subsidy and tax-incentive programs (Boeing, 2016; Chen et al., 2021). I add a structural evaluation of how the allocation of subsidies across the firm-size distribution affects growth and welfare.

The paper proceeds as follows. Section 2 develops the model and derives its main theoretical predictions. Section 3 describes the data, presents the main reduced-form evidence, and provides direct evidence on the process-efficiency channel. Section 4 describes the model's estimation strategy, presents the parameter

estimates, assesses the model's fit, and shows that the process-efficiency mechanism is needed to match the data. Section 5 presents the counterfactual policy experiments and their robustness. Section 6 concludes.

2 Model

My aim in this section is to propose a theory in which a firm's competitiveness on a product line depends on its process efficiency as well as on the quality of its product. That is, I want a model akin to the quality-ladder model of Klette and Kortum (2004), but in which a challenger with a better product must also overcome the incumbent's process efficiency to take over the line. Firms in the model divide their R&D between product innovation and process efficiency, and they are ex ante identical, so they differ only in size.

2.1 Preferences and Production

Consider a closed economy in continuous time. A representative household has constant-relative-risk-aversion preferences over consumption,

$$U = \int_0^\infty e^{-\rho t} \frac{C(t)^{1-\vartheta}}{1-\vartheta}\, dt, \tag{1}$$

with discount rate $\rho > 0$ and relative risk aversion $\vartheta > 0$. The household has a unit measure of members, owns all firms, and supplies its labor inelastically. A fraction L^S of this labor is skilled and is used only in R&D. The rest is unskilled and is used both in assembling the final good and in producing intermediates.

The final good is the numeraire and is produced competitively from unskilled labor and a continuum of intermediate inputs indexed by $j \in [0,1]$,

$$Y = L^\beta \int_{j\in\mathcal{N}} z_j^\beta \, k_j^{1-\beta}\, dj, \tag{2}$$

where L is unskilled labor in final assembly, $\beta \in (0,1)$ is its share of final output, k_j is the quantity of intermediate j, and z_j is its effective productivity. The set of lines that have a producer, the active lines, is $\mathcal{N} \subset [0,1]$, with measure Φ_A.

The effective productivity of line j has two components, one that is specific to the line and one that applies to every line of the firm f that operates it,

$$z_j = q_j\,(1+\psi_f), \tag{3}$$

where q_j is the quality of the line's design and $\psi_f \geq 0$ is the firm's process efficiency. Quality rises only through product innovation. Each intermediate is produced with the linear technology $k_j = \bar{q}\, l_j$, where l_j is the labor input and $\bar{q} \equiv \Phi_A^{-1} \int_{\mathcal{N}} q_j\, dj$ is the average quality of active lines. The marginal cost of every intermediate is therefore $w_u/\bar{q}$. In this specification, process efficiency lowers the cost of a quality-adjusted

unit of the intermediate rather than the cost of a unit of k_j, and this keeps profits linear in $1+\psi_f$.

A line can be produced by more than one firm, namely the incumbent and a challenger who has just developed a new design for it. Following Akcigit and Kerr (2018), I assume that the two compete as follows.

Assumption 1 (Contest for a line) *When a challenger develops a new design for line j, the challenger and the incumbent enter a two-stage game. In the first stage, each pays a fee arbitrarily close to zero. In the second stage, the firms that paid the fee announce their prices.*

Only the firm offering the higher effective productivity can recover the fee in the second stage, so only that firm pays it. The line is therefore served by a single firm, which charges the monopoly price $w_u/[(1-\beta)\,\bar{q}]$. If the challenger loses, its design is never used, so an incumbent with enough process efficiency can keep a better product off the market.

A firm f is characterized by the set $\mathcal{J}_f$ of lines it operates. I denote its portfolio of relative qualities by the multiset $p_f = \{x_j : j \in \mathcal{J}_f\}$, where $x_j \equiv q_j/\bar{q}$, and its aggregate quality by $Q_f \equiv \sum_{j\in\mathcal{J}_f} x_j$. Firms exit at an exogenous Poisson rate $\varphi > 0$, and their lines become inactive. Since profit on a line is proportional to z_j, a firm with aggregate quality Q and process efficiency ψ earns operating profit $\tilde{p}\,(1+\psi)\,Q$ before R&D costs, after normalizing by $\bar{q}$, where $\tilde{p}$ is common to all lines and taken as given.[1] As in Akcigit and Kerr (2018), aggregate quality is therefore the natural measure of firm size.

2.2 Process Efficiency

Firms use skilled labor in two activities, process efficiency and product innovation, which scale differently with firm size and pay off on different horizons. Maintaining process efficiency ψ requires a flow of $c_\psi \psi^\omega$ skilled workers, where $\omega > 1$ governs the convexity of the cost and $c_\psi > 0$ is a cost shifter. The cost does not depend on the number of lines to which ψ is applied, while the benefit $\tilde{p}\,\psi\,Q$ rises with their aggregate quality. This is the cost-spreading logic of Cohen and Klepper (1996). I treat ψ as a static flow choice, so a firm with aggregate quality Q solves

$$\max_{\psi \geq 0} \;\; \tilde{p}\,\psi\,Q \;-\; w_s\,c_\psi\,\psi^\omega, \tag{4}$$

where w_s is the skilled wage.[2] The first-order condition gives

$$\psi^*(Q) = \left[\frac{\tilde{p}\,Q}{\omega\,w_s\,c_\psi}\right]^{1/(\omega-1)}. \tag{5}$$

[1] Online Appendix B derives $\tilde{p}$ in closed form. It depends on assembly labor L, the share β, the share of active lines, and the quality-weighted average of process efficiency across active lines.

[2] Treating ψ as a static choice yields a closed form for $\psi^*(Q)$ and the scalar condition in Assumption 2. In a dynamic formulation, the marginal benefit of ψ would include two additional positive terms: a defense effect, since a higher ψ lowers the rate at which the firm loses lines, and an attack effect, since it raises the firm's own success probability. Both would raise the process efficiency of every incumbent. Online Appendix D contains the derivation.

Hence process efficiency rises with size, at a rate governed by ω. The skilled labor it requires,

$$h_\psi(Q) = c_\psi \left[\psi^*(Q)\right]^\omega = c_\psi \left[\frac{\tilde{p}\, Q}{\omega\, w_s\, c_\psi}\right]^\eta, \qquad \eta \equiv \frac{\omega}{\omega - 1} > 1, \tag{6}$$

is increasing and convex in Q. Net of this cost, the flow profit of the firm is

$$\pi(Q) = \tilde{p}\, Q + \hat{D}\, Q^\eta, \qquad \text{where} \quad \hat{D} \equiv \frac{\omega - 1}{\omega}\, \tilde{p}^{\,\eta}\, (\omega\, w_s\, c_\psi)^{-1/(\omega-1)}. \tag{7}$$

The first term is the profit the firm would earn with $\psi = 0$, which is its entire flow profit in the Klette–Kortum model. The second term is the net surplus from process R&D. It is superlinear in Q because a larger firm both chooses a higher $\psi^*(Q)$ and applies it to more lines. Hence, unlike in Klette and Kortum (2004) and Akcigit and Kerr (2018), firm value is not linear in the qualities of the firm's lines.

2.3 Product Innovation and Blocking

A firm that employs h_X skilled workers in product R&D generates innovation attempts at the Poisson rate

$$X = \theta\, h_X^{1-\gamma}, \tag{8}$$

where $\theta > 0$ is innovation productivity, common to all firms, and $\gamma \in (0, 1)$ captures decreasing returns within the R&D lab. The rate does not depend on the number of lines the firm owns.[3] Each attempt targets a line drawn at random from $[0, 1]$. When an attempt on an active line succeeds, the innovator takes over the line and its quality improves from q_j to $(1 + \lambda)\, q_j$, where $\lambda > 0$ is the quality step. An attempt on an inactive line always succeeds. The line then re-enters the economy at quality $(1 + \lambda)\, \bar{q}$, so the innovation builds on the average quality in the economy.

A unit measure of potential entrants generates attempts at the aggregate rate X_e by hiring $h_e = c_e\, X_e^{1+\alpha_e}$ skilled workers, where $c_e > 0$ and $\alpha_e > 0$. The convexity of this cost makes entry a smooth function of the skilled wage. A successful entrant becomes a firm with one line. Since an entrant has no lines over which to spread a process improvement, it attempts with $\psi = 0$.

By Assumption 1, an attempt by firm f on a line held by firm f' succeeds only if the challenger offers the higher effective productivity, $(1 + \lambda)\, q_j\, (1 + \psi_f) > q_j\, (1 + \psi_{f'})$, namely,

$$(1 + \lambda)(1 + \psi_f) > 1 + \psi_{f'}. \tag{9}$$

Because the step is proportional, q_j cancels, and the condition does not depend on the quality of the line. A successful challenger brings its own process efficiency to the line, while the process efficiency of the incumbent

[3] In Klette and Kortum (2004), the attempt rate is homogeneous of degree one in R&D and the number of lines, so that a firm's past innovations make it more innovative. As in the baseline model of Akcigit and Kerr (2018), I shut down this dependence, so that larger firms have no advantage in generating attempts. Their advantage lies instead in making attempts succeed, and it comes from process efficiency.

applies to its remaining lines. If the condition fails, the line keeps its quality q_j.

For an entrant, (9) reduces to $\lambda > \psi_{f'}$. Since $\psi^*(Q)$ is increasing, there is an aggregate quality

$$\bar{Q}_e \equiv \frac{\omega\, w_s\, c_\psi}{\tilde{p}}\, \lambda^{\omega-1} \tag{10}$$

above which no entrant can displace the incumbent, however much it spends on R&D.

Suppose that firms are distributed over portfolios according to a measure M, with total mass F, and that active lines are distributed according to S over pairs (x, ψ'), where x is the relative quality of a line and ψ' the process efficiency of the firm that holds it. Both distributions are determined in equilibrium, and firms take them as given. Denote by $\psi(p)$ the process efficiency of a firm with portfolio p, so that $\psi(p) = \psi^*(Q(p))$. Condition (9) is a step function, which I replace with the logistic function $\Lambda(A) = [1+\exp(-\kappa A)]^{-1}$, $\kappa > 0$, where $A(\psi, \psi') \equiv (1+\lambda)(1+\psi) - (1+\psi')$ is the advantage of a challenger with process efficiency ψ over an incumbent with process efficiency ψ'.[4] An attempt by a firm with process efficiency ψ then succeeds with probability

$$B(\psi) = \Phi_I + \Phi_A \int \Lambda\big(A(\psi, \psi')\big)\, dS(x, \psi'), \tag{11}$$

where $\Phi_I = 1 - \Phi_A$ is the share of inactive lines, and an entry attempt succeeds with probability $B_e = B(0)$. Since Λ rises with the challenger's process efficiency, B increases with ψ and hence with Q. That is, for the same R&D, a larger firm captures more lines.

The same contest determines how fast incumbents lose lines. A line held by a firm with process efficiency ψ receives attempts from entrants at rate X_e and from firms with portfolio p'' at rate $X^*(p'')\, dM(p'')$. Hence it is displaced at rate

$$\tau(\psi) = X_e\, \Lambda\big(A(0, \psi)\big) + \int X^*(p'')\, \Lambda\big(A(\psi(p''), \psi)\big)\, dM(p''). \tag{12}$$

2.4 Firm Value and Entry

Unlike ψ, the attempt rate X is a dynamic choice, because a line the firm captures pays a flow profit until it is taken over. Consider a firm with portfolio p, which serves as the state variable in the firm's problem, and let $\hat{V}(p)$ denote its value normalized by $\bar{q}$, with $\hat{V}(\emptyset) = 0$. The term $p \oplus y$ denotes the portfolio p with a line of relative quality y added, and $p \ominus x_j$ the portfolio without line j. For any function f of portfolios, let

$$\mathcal{G}f(p) \equiv \Phi_I \big[f(p \oplus (1+\lambda)) - f(p)\big] + \Phi_A \int \Lambda\big(A(\psi(p), \psi')\big) \big[f(p \oplus (1+\lambda)x) - f(p)\big]\, dS(x, \psi') \tag{13}$$

denote the expected change in f from one attempt. The first term corresponds to an attempt on an inactive line, while the second denotes the capture of an active line of relative quality x. The expected gain from an attempt is then $G(p) \equiv \mathcal{G}\hat{V}(p)$.

[4] The smoother is adopted for computational convenience. As $\kappa \to \infty$, Λ converges to the step function, and a finite κ smooths the transition without changing the qualitative results.

On the balanced growth path, the household's Euler equation gives the interest rate $r = \rho + \vartheta g$, where g is the growth rate of $\bar{q}$, and firm value satisfies

$$\hat{r}\,\hat{V}(p) = \max_{X \geq 0} \left\{ \pi(Q(p)) - w_s \left[\frac{X}{\theta}\right]^{1/(1-\gamma)} + X\,G(p) \right\} - \tau(\psi(p)) \sum_j \left[\hat{V}(p) - \hat{V}(p \ominus x_j)\right] - g \sum_j x_j \,\frac{\partial \hat{V}(p)}{\partial x_j}, \quad (14)$$

where $\hat{r} = \rho + (\vartheta - 1)g + \varphi$. The discount rate subtracts g because $\hat{V}$ is measured in units of $\bar{q}$, which grows at rate g, and adds φ because the firm exits at that rate. The first term on the right-hand side is flow profit net of product R&D plus the expected gain from attempts. The second is the expected loss from creative destruction, where $\hat{V}(p \ominus x_j)$ is firm value after the loss of line j. The last term is the erosion of the firm's relative quality as average quality grows. The first-order condition equates the marginal cost of an attempt to its expected gain,

$$\frac{w_s}{(1-\gamma)\,\theta} \left[\frac{X^*(p)}{\theta}\right]^{\gamma/(1-\gamma)} = G(p). \quad (15)$$

Product R&D labor is then $h_X(p) = [X^*(p)/\theta]^{1/(1-\gamma)}$.

Entrants hold empty portfolios, so the expected value of an entry attempt is $G(\emptyset) = \Phi_I\,\hat{V}(\{1+\lambda\}) + \Phi_A \int \Lambda(\lambda - \psi')\,\hat{V}(\{(1+\lambda)x\})\,dS(x,\psi')$. Potential entrants choose X_e to maximize $X_e\,G(\emptyset) - w_s c_e X_e^{1+\alpha_e}$, which gives

$$G(\emptyset) = c_e\,(1+\alpha_e)\,X_e^{\alpha_e}\,w_s. \quad (16)$$

2.5 Balanced Growth Path

I focus on a balanced growth path on which all variables normalized by $\bar{q}$ are constant and the distributions M and S are stationary. While successful attempts, line losses, exit, and growth in average quality move individual portfolios around, the measure of firms over portfolios is stable. That is, for every function f of portfolios with $f(\emptyset) = 0$,

$$\int \Big\{ X^*(p)\,\mathcal{G}f(p) - \tau(\psi(p)) \sum_j \left[f(p) - f(p \ominus x_j)\right] - g \sum_j x_j \,\frac{\partial f(p)}{\partial x_j} - \varphi\, f(p) \Big\}\, dM(p) + X_e\,\mathcal{G}f(\emptyset) = 0, \quad (17)$$

where the term in braces is the expected rate of change of $f(p)$ for a firm with portfolio p and the last term is the inflow of entrants. The distribution of lines S is the one implied by M. In particular, the measure of active lines is $\Phi_A = \int n(p)\,dM(p)$, where $n(p)$ is the number of lines in p, and $\int x\,dS = 1$ by the definition of $\bar{q}$.

Attempts arrive on each line at the aggregate attempt rate $\bar{\tau} \equiv X_e + \int X^*(p)\,dM(p)$, so the share of active lines is $\Phi_A = \bar{\tau}/(\bar{\tau} + \varphi)$. The skilled wage clears the market for skilled labor, which is used in entry, product, and process R&D,

$$h_e + \int h_X(p)\,dM(p) + \int h_\psi(p)\,dM(p) = L^S, \quad (18)$$

where $h_\psi(p) = c_\psi\,\psi(p)^\omega$.

Finally, the growth rate is determined by the frequency of successful innovations and their size. A successful attempt on an active line of relative quality x adds $\lambda x\,\bar{q}$ to aggregate quality, and a line that re-enters the economy adds $(1+\lambda)\,\bar{q}$, while an exiting line removes $\bar{q}$ on average, since exit is unrelated to quality. Since lines re-enter and exit at the same rate on the balanced growth path, $\bar{\tau}\,\Phi_I = \varphi\,\Phi_A$, average quality grows at rate

$$g = \lambda\left[\int x\,\tau(\psi')\,dS(x,\psi') + \varphi\right]. \tag{19}$$

Blocking therefore lowers growth by lowering τ, and the effect is largest on the lines of firms with high process efficiency. To the extent that these firms hold the highest-quality lines, a blocked step costs more growth than an average one.

Definition 1 (Balanced growth path) *A balanced growth path consists of a skilled wage w_s, policies $\psi(p)$ and $X^*(p)$, values $\hat{V}(p)$, an entry rate X_e, a measure of firms M with induced line distribution S, a share of active lines Φ_A, and a growth rate g such that (i) $\psi(p) = \psi^*(Q(p))$ satisfies (5); (ii) $\hat{V}$ and X^* satisfy (14) and (15); (iii) X_e satisfies (16); (iv) M satisfies (17); (v) $\Phi_A = \bar{\tau}/(\bar{\tau}+\varphi)$, and (18) and (19) hold; and (vi) $\int x^\eta\,dS < \infty$.*[5]

2.6 Firm Size, Blocking, and Subsidies

All proofs are relegated to Online Appendix C.

Proposition 1 (Process efficiency insulates large incumbents) *(a) The displacement rate $\tau(\psi^*(Q))$ of a firm's lines is decreasing in Q and converges to zero as $Q \to \infty$. (b) In the step-function limit $\kappa \to \infty$, no entrant can displace a firm with $Q > \bar{Q}_e$, and the set of incumbents that can displace a firm shrinks as Q rises.*

Note that Proposition 1 is stated in terms of aggregate quality Q rather than the number of lines, so a firm with a few high-quality lines can be better protected than one with many low-quality lines. As Q rises, fewer firms are able to take the firm's lines: entrants are the first to be excluded, and small incumbents follow. This protection may erode over time. As average quality in the economy rises at rate g, the relative quality of the firm's lines gradually declines if the firm does not innovate, and hence Q falls. The firm's process efficiency then falls, and it becomes easier to displace.

Now consider how firms divide their R&D. Define the share of a firm's R&D labor devoted to process efficiency as

$$\chi(p) \equiv \frac{h_\psi(p)}{h_\psi(p) + h_X(p)}. \tag{20}$$

[5]Condition (vi) ensures that the expected value of an attempt is finite, since flow profit is of order Q^η and a line of high relative quality x is therefore worth an amount of order x^η. The distribution of relative quality is stationary because lines leave the economy at rate φ and re-enter it at quality $(1+\lambda)\,\bar{q}$.

Comparing how the two components grow with size requires one parameter restriction, without which product R&D labor can grow with firm size as quickly as process R&D labor.

Assumption 2 $\omega > 1/\gamma$.

Proposition 2 (Larger firms devote more R&D labor to process efficiency) *Under Assumption 2, $\chi(p) \to 1$ as $Q(p) \to \infty$, regardless of the composition of the firm's portfolio.*

The marginal return to process efficiency is $\tilde{p}\,Q$, since every line benefits from the same improvement, while product innovation faces decreasing returns at the level of the firm. For large Q, h_ψ grows as $Q^{\omega/(\omega-1)}$ and h_X as $Q^{1/[\gamma(\omega-1)]}$, and the former exponent is larger if and only if $\omega > 1/\gamma$. The composition of the portfolio does not matter in the limit, because a large firm rarely loses lines.

Now consider a size-contingent subsidy on the cost of R&D labor. A firm with portfolio p faces the skilled wage $(1-\sigma(p))\,w_s$ on both product and process R&D, and entrants face $(1-\sigma_e)\,w_s$. The rate $\sigma(p) \in [0,1)$ depends on a measure of size that rises when the firm adds a line, such as its number of lines or its employment. The subsidies are financed by a lump-sum tax on the household. Because the composition of R&D is not contractible, the firm decides for itself how to divide the subsidized labor between the two activities. Its process efficiency is given by (5) with w_s replaced by $(1-\sigma(p))\,w_s$, and its attempt rate rises through (15). I restrict attention to schedules $\sigma(\cdot)$ that are weakly increasing in size.[6] A *subsidy to large firms* raises $\sigma(p)$ for every firm above some size and leaves it unchanged for all smaller firms and for entrants.

Proposition 3 (Subsidies to large firms fund process efficiency) *Consider a marginal increase in the subsidy rate of a firm with portfolio p, starting from no subsidy and holding fixed w_s, the firm's success probabilities, and $\hat{V}(\cdot)$. Then the share of the additional skilled labor that the firm allocates to process efficiency is*

$$\varsigma(p) = \frac{\gamma\eta\,\chi(p)}{\gamma\eta\,\chi(p) + 1 - \chi(p)},$$

which is increasing in $\chi(p)$. If, in addition, Assumption 2 holds, then $\varsigma(p) \to 1$ as $Q(p) \to \infty$.

That is, a large firm spends most of a subsidy on process efficiency, which makes its lines harder for other firms to take.

Proposition 4 (Subsidies to large firms discourage innovation by smaller firms) *Consider a subsidy to large firms. (a) Given the distribution of firms, the success probability of every smaller firm and of every entrant falls. (b) Given, in addition, the skilled wage and firm values, the attempt rate of every smaller firm and of entrants falls.*

Proposition 4 implies that, holding fixed their own subsidies, small firms innovate less, and hence file fewer patents, when a larger share of subsidies goes to large firms. Also note that part (b) of Proposition 4

[6] The restriction is sufficient for part (b) of Proposition 4, which requires only that firm value rise with each additional line (Lemma C.2 in Online Appendix C).

holds the skilled wage and firm values fixed. In general equilibrium, a subsidy to large firms also raises their attempt rates and the skilled wage, so its net effect on growth and welfare has to be determined quantitatively.

The model has two sources of inefficiency. The first is standard in quality-ladder models. An innovator chooses the rate of product innovation based only on its profits, and does not internalize the quality improvement that future innovators build on. The equilibrium rate of creative destruction can therefore be too low relative to the social optimum. The second is specific to this model. A firm chooses ψ to raise the profit on its own lines and ignores that a higher ψ lowers the success probability of every challenger. The loss accrues in the future, since at the time of a blocked attempt the line stays with the more productive firm, regardless of the source of that productivity. In the future, however, the economy loses the forgone quality improvement, since improvements in quality persist across periods and displacements, whereas process efficiency is specific to the firm and leaves the line once the firm loses it. Hence, in equilibrium, process efficiency can be overprovided. A policy that could distinguish between the two activities would shift skilled labor from process efficiency toward product innovation. Since such a policy is not available, a feasible alternative is to condition subsidies on firm size, which predicts the composition of R&D.

3 Empirical Evidence

This section tests the model's central prediction, that concentrating subsidies at large firms crowds out patenting by small firms. I describe the data, present the reduced-form evidence, and then report survey evidence corroborating the model's mechanism that process R&D is more prevalent at large firms.

3.1 Data

I employ the firm-level panel from the Chinese Tax Survey (CTS) covering the years 2008–2020. The CTS is administered by the State Administration of Taxation, China's tax authority. It covers firms across all sectors and records annual information on employment, revenue, R&D expenditure, and fiscal subsidies received from government programs, along with a broad set of tax and balance-sheet items.[7] The subsidy variable records direct fiscal transfers from government programs such as R&D grants, innovation-fund awards, and High-Technology Enterprise (HTE) disbursements. It does not capture implicit forms of support like below-market credit, preferential land grants, or tax-rate reductions. Therefore, the subsidy totals used throughout the paper can be seen as a lower bound on aggregate government support.

The CTS sample is stratified by firm size, and firms above certain revenue thresholds are sampled with higher probability. For the structural moments used in estimation, I re-weight CTS observations by an inverse-probability scheme that matches the bin shares of the Chinese National Economic Census, which is

[7]The CTS is distinct from the Chinese Industrial Enterprise Survey (CIES), which is more frequently used in the macro-development literature. The CIES only covers state-owned and above-scale industrial firms, whereas the CTS samples all sectors with strata defined by firm size. See Chen et al. (2021) for the sampling design. The panel splices two extracts with different time frames, and coverage falls from roughly 720,000 observations a year in 2008–2015 to 455,000 in 2017–2020.

intended to cover all non-agricultural firms in China. To restrict my analysis to private-sector firms, I drop all observations representing state-owned and state-controlled enterprises, government bodies, and public institutions (e.g., universities and hospitals), identifying these entities from their registration codes.

I then merge the firm-level panel with patent application records from the China National Intellectual Property Administration (CNIPA). The records give the date and type (i.e., invention patent, utility model, or design patent) of each application filed by each firm. Following standard practice, I take invention patents as the measure that is most closely associated with genuine technological advance. Throughout the paper, a firm's patents are its granted invention patents in a year, and its R&D is the deductible R&D expense on its annual tax filing.[8]

The merged panel contains approximately 8.2 million firm-year observations. I then further restrict the analysis to firms with at least one year of positive R&D expenditure, because the model's mechanism operates through firms that do R&D. This leaves a working sample of approximately 1.9 million firm-year observations.[9]

3.2 Reduced-Form Evidence

Empirical design. In the model, firms draw innovation targets from a single pool of product lines. For the empirical strategy I take that pool as an industry-year. Within each industry-year cell, I ask whether a larger share of the cell's subsidies going to its largest firms lowers patenting by smaller firms, holding each group's own subsidy intensity fixed so that the coefficient reflects the spillover from concentration rather than the direct effect of a group's own receipts. I use cell fixed effects to control for all shocks that affect the size groups uniformly.

The unit of observation is a 3-digit industry × year × size-group.[10] I partition firms by employment into Small (≤ 35), Medium (36–395), and Large (> 395) groups, with cutoffs corresponding to the 35th and 85th percentiles of the employment distribution in the working sample. In each industry-year cell jt, subsidy concentration is the share of total subsidies flowing to the top 10% firms in that cell by employment:

$$\text{Conc}_{jt}^{90} \;=\; \frac{\sum_{i:\,\text{emp}_{ijt} > q_{jt}} \text{Subsidy}_{ijt}}{\sum_i \text{Subsidy}_{ijt}}, \tag{21}$$

[8] The firm-patent merge uses exact-string equality between the CTS firm name and the CNIPA applicant field. When a patent has several applicants, each listed applicant is credited with one patent. Exact matching yields a match rate of 19.85% over 2008–2020. Non-matches reflect both applicant-name variations that exact matching cannot identify (e.g., differences in corporate-form suffixes, punctuation, or abbreviation) and patents filed by firms outside the CTS sampling frame. I experimented with three fuzzy matching methods: Jieba-tokenized fuzzy matching, a hybrid token-and-character similarity measure, and RapidFuzz's token-sort ratio. They raise coverage by 1.5 to 10 percentage points but produce many false positives. Between 15% and 57% of fuzzy matches map a single patent to multiple candidate firms. I therefore use exact-string matching, which means that reported patent counts are weakly underestimated.

[9] I also trim observations above the 99.99-th percentile of employment, revenue, and assets to limit the influence of reporting errors in the extreme right tail. The trim removes only a few hundred observations.

[10] The 3-digit level corresponds to industry classes in China's GB/T 4754 national industry classification standard. The working sample covers 263 such classes. Examples of such classes are pharmaceutical preparations, automobile manufacturing, and electronic component manufacturing.

where q_{jt} is the 90th percentile of employment in the cell. The estimating equation is

$$\begin{aligned}\log(1+\text{Patents}_{jt,g}) \;=\; & \alpha_{jt} + \beta_S\,\text{Conc}^{90}_{jt}\times\mathbf{1}[g{=}S] + \beta_L\,\text{Conc}^{90}_{jt}\times\mathbf{1}[g{=}L] \\ & + \phi_S\,\text{RevConc}^{90}_{jt}\times\mathbf{1}[g{=}S] + \phi_L\,\text{RevConc}^{90}_{jt}\times\mathbf{1}[g{=}L] \\ & + \mathbf{X}'_{jt,g}\boldsymbol{\gamma} + \varepsilon_{jt,g},\end{aligned} \tag{22}$$

where α_{jt} is the 3-digit industry × year cell fixed effect, $\mathbf{X}_{jt,g}$ collects the size group's own subsidy intensity and the log of its firm count, RevConc^{90}_{jt} is defined analogously to equation (21) with revenue in place of subsidies, and Medium ($g{=}M$) is the omitted reference category. The coefficients of interest are β_S and β_L, which measure how an increase in subsidy concentration within the cell affects patenting in each size group relative to medium firms in the same industry-year. The revenue-concentration terms are included because subsidy concentration tracks revenue concentration: in industry-years where a few large firms account for most revenue, even a policy that distributes funds in proportion to scale would produce a high Conc^{90}_{jt}. Without these terms, β_S and β_L would pick up market structure rather than subsidy policy. The choice of a top 10% cutoff for subsidy and revenue concentration is arbitrary, but the results hold with alternative cutoffs from the top 1% to the top 50%.[11] The estimating sample has 5,664 cell-group observations from 263 3-digit industries over 2008–2020.

Subsidy concentration may be endogenous. If officials direct funds toward firms that are already growing, industries with strong small-firm prospects would attract less concentrated subsidies, and OLS would overstate $|\beta_S|$. Controlling for cell fixed effects would not help, because the unobservables may affect the size groups asymmetrically. The next subsection addresses this with a shift-share instrument.

Shift-share identification. In China, innovation subsidies are administered at the provincial level, and provinces differ persistently in their tendency to favor large firms over smaller ones. These differences are often slow-moving and may originate from diverse leadership priorities and industrial-policy orientation. An industry whose initial employment is concentrated in provinces that favor large firms will therefore show higher subsidy concentration than an industry located elsewhere, for reasons unrelated to its own innovation prospects. This motivates a shift-share instrument for $\text{Conc}^{90}_{jt}\times\mathbf{1}[g]$ that interacts baseline employment shares with each province's tilt of subsidies toward large firms. Formally,

$$B_{jt} \;=\; \sum_{p} z_{jp,2008} \;\times\; \text{LargeBias}_{p,-j,t}, \tag{23}$$

where $z_{jp,2008}$ is the share of industry j's 2008 employment located in province p and $\text{LargeBias}_{p,-j,t}$ is the share of province p's innovation subsidies received by large firms in year t, computed across all industries except j. The leave-out construction keeps industry j's own subsidies from entering the instrument through

[11] Online Appendix Table F.1 reports equation (22) estimated at the top 1%, 5%, 25%, and 50% cutoffs. The small-firm coefficient β_S is negative and significant under both OLS and the Bartik IV at every cutoff.

Table I: The effect of subsidy concentration on patenting by size group

	(1) *Cell FE + Levels*	(2) *+ RevConc × group*	(3) *+ Firm Comp.*	(4) *4d Cell FE (tighter)*
Panel A: OLS				
$\text{Conc}^{90} \times$ Small	-3.584^{***}	-0.675^{***}	-0.682^{***}	-0.557^{***}
	(0.101)	(0.176)	(0.171)	(0.100)
$\text{Conc}^{90} \times$ Large	2.802^{***}	0.242	0.240	0.215^{**}
	(0.113)	(0.194)	(0.189)	(0.105)
Panel B: Bartik IV				
$\text{Conc}^{90} \times$ Small	-4.168^{***}	-1.547^{***}	-1.559^{***}	-1.093^{***}
	(0.171)	(0.337)	(0.343)	(0.190)
$\text{Conc}^{90} \times$ Large	3.350^{***}	0.376	0.283	0.271
	(0.185)	(0.236)	(0.239)	(0.184)
First-stage F (Small / Large)	7,374 / 6,972	95 / 92	95 / 89	195 / 175
Cell × Group Controls	Yes	Yes	Yes	Yes
$\text{RevConc}^{90} \times$ group	No	Yes	Yes	Yes
Group High-Tech composition	No	No	Yes	Yes
Cell fixed effects	3d × yr	3d × yr	3d × yr	4d × yr
Observations	5,664	5,664	5,664	13,431

Notes: Dependent variable is $\log(1 + \text{Patents}_{jt,g})$ in a 3-digit industry × year × size-group cell, with Medium as the reference group. All columns include the group's own subsidy intensity and the log of its firm count. Column 1 has cell fixed effects only; columns 2–4 add, in turn, $\text{RevConc}^{90}_{jt} \times \mathbf{1}[g]$, the group's high-tech firm share, and 4-digit industry × year fixed effects. Panel B instruments $\text{Conc}^{90}_{jt} \times \mathbf{1}[g]$ with the shift-share in equation (23). Standard errors clustered by 3-digit industry in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the province-level aggregates.

Identification rests on two assumptions. *Share exogeneity* requires that baseline (2008) employment shares be uncorrelated with future industry-year-specific patenting trends across size groups (Goldsmith-Pinkham et al., 2020). *Shock exogeneity* requires that the leave-out provincial bias be orthogonal to industry j's own innovation shocks (Borusyak et al., 2025). The diagnostics in Online Appendix F offer support for both assumptions. First-stage partial F-statistics range from 89 to over 7,000 across the columns of Table I. A changes-based pre-trend test is insignificant for the small-firm interaction.

Results. Table I reports four specifications of equation (22), with OLS in Panel A and Bartik IV in Panel B. The columns progressively tighten the specification: column 1 has only the cell fixed effect and the group-level controls in $\mathbf{X}_{jt,g}$; column 2 adds the revenue-concentration × group interactions; column 3 adds the within-group share of HTE-certified firms, which enjoy preferential subsidy access and are systematically more R&D-intensive than non-certified peers in the same size group; and column 4 tightens the cell fixed

effect to 4-digit industry × year.[12]

The small-firm coefficient is negative and significant in every column under both estimators. With the revenue-concentration interactions added (column 2 onward), the OLS estimates lie between −0.56 and −0.68 ($p < 0.01$), and the IV estimates between −1.09 and −1.56 ($p < 0.01$). The IV coefficients are about twice the OLS coefficients, possibly because of the high correlation between subsidy and revenue concentration. Partialing out revenue concentration leaves little true variation in subsidy concentration, so measurement error in subsidy concentration attenuates OLS more severely. I therefore interpret the OLS coefficient as a lower bound on the true effect. Meanwhile, the large-firm coefficient is insignificant from column 2 onward. This shows that increasing subsidy concentration in an industry-year has little effect on large-firm patenting, conditional on revenue concentration and the *levels* of the large firms' subsidy receipt.

Column 3 is the preferred specification. A 10-percentage-point increase in the top-decile subsidy share is associated with a 0.068-log-point (roughly 7%) reduction in small-firm patenting relative to medium firms in the same industry-year. The more conservative 4-digit specification in column 4 implies a reduction of roughly 5%.[13] This aligns with what the model predicts. By Proposition 3, subsidies at large firms are mostly allocated to process R&D, which raises the bar that smaller challengers must clear to displace them, crowding out small-firm patenting.

Note that the dependent variable I use is the number of invention patents firms file in an industry-year. Firms may, however, file patents strategically to qualify for tax preferences, subsidies, and procurement eligibility. If that is the case, a decline in small-firm patenting need not reflect a decline in inventive activity. I therefore restrict the dependent variable to *breakthrough patents*, the top 10% of patents each year by a text-based inventiveness measure adapted from Kelly et al. (2021) to Chinese patent text.[14] Strategic filings should rarely rank in the top decile of inventiveness, yet Online Appendix Table F.2 shows that the negative small-firm coefficient remains economically large and statistically significant. It also stays negative and significant across concentration cutoffs from the median to the 99th percentile, under two-way clustering on industry and year, and after excluding the top-1% subsidy recipients (Online Appendix Tables F.1, F.3 and F.4).

3.3 Additional Empirical Evidence

So far I have documented the crowding-out channel that the model predicts. I now check two features underlying it, that large firms devote more of their R&D to process improvement (Proposition 2) and that the output of process R&D is protected through secrecy rather than patents, against aggregate tabulations from the National Enterprise Innovation Survey, a census of above-scale firms conducted by the National

[12] The 4-digit level subdivides each 3-digit class into narrower subclasses. Within automobile manufacturing, for instance, the 4-digit subclasses include complete-vehicle, modified-vehicle, special-purpose vehicle, and auto-parts manufacturing. The estimating sample covers 13,431 observations across 803 subclasses.

[13] In the estimating sample, Conc_{jt}^{90} averages 0.44 with a 10th–90th percentile range of 0.15 to 0.74, so the 10-percentage-point illustration is well within typical between-cell variation.

[14] Online Appendix E offers details of the measure's construction.

Bureau of Statistics and the Ministry of Science and Technology.[15] Among industrial firms conducting product innovation, the share that also conducts process innovation rises monotonically with size, from 65.0% of micro firms to 71.3% of small, 80.0% of medium, and 87.1% of large firms, and a parallel survey of below-scale firms extends the gradient downward, with only 55.4% of below-scale innovators engaging in process innovation. This is what cost-spreading predicts: among firms already doing product R&D, the incentive to add process improvement scales with the number of product lines over which the cost reduction amortizes. Trade-secret adoption follows the same gradient, from 8.4% of micro firms to 16.8% of small, 25.3% of medium, and 37.0% of large firms, and large firms are four times more likely than micro firms to cite imitation risk as a principal barrier to innovation (9.3% vs. 2.3%), consistent with firms favoring secrecy over patents when imitation risk is high.

4 Estimation and Quantitative Analysis

This section estimates the model by simulated method of moments, assesses its fit, and shows that without process efficiency the model cannot match the data.

4.1 Identification

Externally calibrated parameters. Seven parameters are set externally. The discount rate $\rho = 0.02$, the inverse intertemporal elasticity of substitution $\vartheta = 2$, and the unskilled-labor share $\beta = 0.33$ take standard values. The skilled labor share $L^S = 0.161$ equals the share of the Chinese workforce with science and engineering training, measured in the 2010 Population Census, and the smoothing parameter $\kappa = 10$ governs the logistic approximation used for the innovation success rate.

The curvature of the R&D production function $\gamma = 0.5$ follows microeconometric estimates of the elasticity of patents to R&D: the survey of Griliches (1990) finds a range of 0.3–0.6, U.S. count-data studies (Blundell et al., 2002) find values close to 0.5, and Chinese firm data (Wu et al., 2022) find the same range. Acemoglu et al. (2018) adopt the same value on these grounds. The exogenous exit rate $\varphi = 0.004$ is calibrated to the rate at which large continuing firms in the panel exit for reasons unrelated to product-market competition.[16]

[15] I use the 2023 yearbook (2022 survey year). The survey reports aggregate statistics by firm-size stratum (large, medium, small, micro, per the 2017 NBS classification standard). Size categories are defined jointly by employment and revenue with industry-specific thresholds.

[16] I compute this rate from firms in the panel with employment above 785, which correspond to the top 0.1% firms by employment in the model at the estimated parameters. For these firms, the probability of exit from having all product lines stolen in a single period is negligible. In the data, I classify each firm as exiting in year t if it ceases to appear in the tax records and does not reappear before 2020. Using this approach, the implied annual exit hazard in the sample is approximately 0.4 percent.

Simulated method of moments. The estimator minimizes the sum of squared percentage deviations between model and data moments,

$$\hat{\Theta} \;=\; \arg\min_{\Theta} \sum_{k=1}^{7} \left(\frac{m_k(\Theta) - m_k^d}{|m_k^d|} \right)^2, \tag{24}$$

where m_k^d is the data value of the k-th moment and $m_k(\Theta)$ its model counterpart at parameter vector Θ. I assign equal weight to all moments. Online Appendix I describes the numerical solution of the balanced growth path that delivers $m_k(\Theta)$. Table III lists the seven targeted moments I use. All moments are computed from the 2016–2020 R&D-active panel of 52,091 firms, with the observations re-weighted such that the share of each firm-size bin matches its share in the Chinese National Economic Census. There are six estimated parameters: the process-efficiency cost convexity ω, its scale c_ψ, the incumbent innovation capacity θ, the quality step λ, the entry-cost convexity α_e, and the entry-cost level H, defined as the entry cost evaluated at a fixed reference attempt rate.[17]. The system is overidentified by one degree of freedom.

Identification heuristics. Below I offer some heuristics of how each parameter is identified. The quality step λ is pinned down by aggregate growth: given the equilibrium rate of creative destruction, equation (19) maps growth into the size of the step. The two entry-cost parameters are pinned down by the three entry moments. The level H is identified from the entry rate, the share of firms in their first year of positive R&D, and the employment share of those firms, through the entry condition (16). The convexity α_e is disciplined by the share of R&D performed by first-year firms, because a steeper marginal entry cost limits how much skilled labor entry attempts can absorb. The two process-efficiency parameters are pinned down by the size profiles of R&D intensity and of employment. The convexity ω is identified from the slope of log(R&D/Rev) on log employment, since a higher ω flattens the size profile of ψ^* in (5) and hence the slope. Given ω, the scale c_ψ is pinned down by the variance of log employment, since the level of process investment governs how dispersed $Q\,(1 + \psi^*(Q))$ is across firms. The innovation capacity θ is pinned down by the slope of log(Pat/R&D) on log employment. Patents per unit of R&D fall with size as larger firms tilt their R&D toward process efficiency; given ω, the rate of that decline pins down θ, which moves product innovation up or down at every size.

Standard errors come from the GMM sandwich formula, with the covariance of the data moments obtained by a firm-cluster bootstrap. Online Appendix G gives the details.

4.2 Benchmark Estimation Results

This subsection reports the parameter estimates, the model's fit to targeted and nontargeted moments, and the properties of the calibrated economy.

[17] $H \equiv c_e \cdot X_0^{1+\alpha_e}$ at the reference rate $X_0 = 0.01$. I report H rather than c_e because the units of c_e depend on α_e through the entry cost $h_e = c_e X_e^{1+\alpha_e}$, which makes c_e difficult to compare across specifications with different $\hat{\alpha}_e$. The choice of X_0 only rescales the reported level and does not affect estimation.

Table II: Parameter estimates

Parameter	Description	Value	S.E.
ω	Process eff. cost convexity	3.279	0.961
c_ψ	Process eff. cost scale	0.323	0.031
θ	Incumbent innov. capacity	0.149	0.005
λ	Quality step size	1.079	0.095
α_e	Entry-cost convexity	2.987	1.702
H	Entry-cost level	0.527	0.182

Notes: Standard errors are computed from the GMM sandwich formula described in Online Appendix G.

Parameter estimates. The estimated process-efficiency cost convexity is $\hat{\omega} = 3.28$, with a standard error of 0.96. It lies above the threshold $1/\gamma = 2$ of Assumption 2, so the composition mechanism of Proposition 2 is active at the point estimates. The quality step $\hat{\lambda} = 1.08$ implies that a successful product innovation roughly doubles a line's quality. The incumbent innovation capacity $\hat{\theta} = 0.15$ sets the level of the product-innovation production function, and the entry-cost parameters are $\hat{H} = 0.53$ and $\hat{\alpha}_e = 2.99$, with a larger standard error for $\hat{\alpha}_e$ than for the other parameters.

Table III: Model and data moments

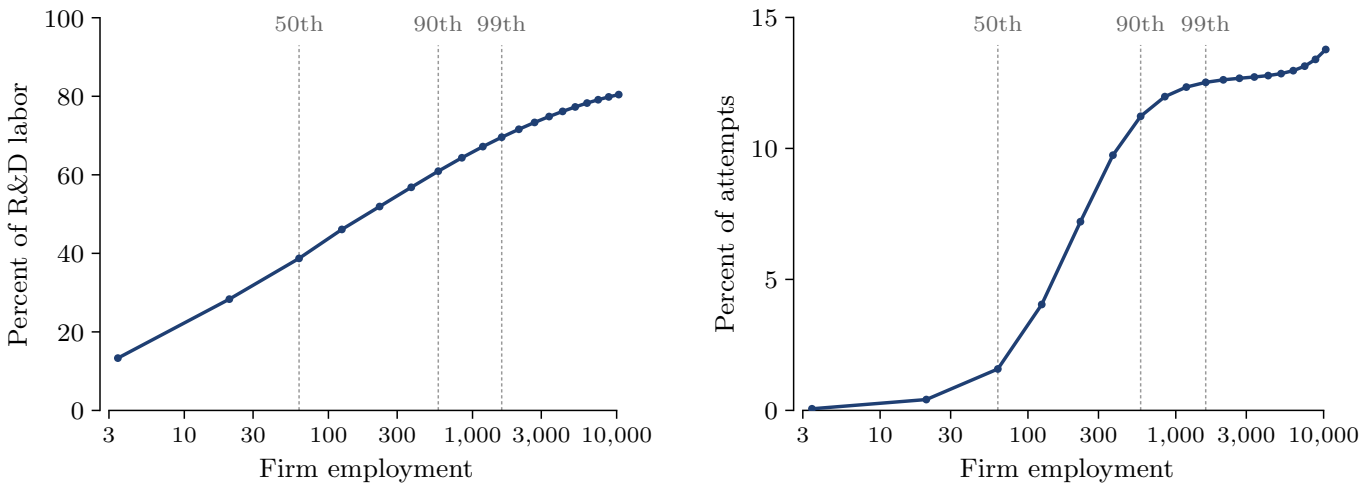
Moment	Model	Data	S.E.
Aggregate growth	0.0231	0.0222	0.0014
Entry rate	0.0132	0.0114	0.0011
Entrant employment share	0.0028	0.0031	0.0008
log(R&D/Rev) vs size	−0.5225	−0.6141	0.0072
log(Pat/R&D) vs size	−0.5466	−0.5606	0.0160
Entry R&D share	0.0204	0.0190	0.0037
Var(log Emp)	0.91	1.27	0.0713

Notes: Standard errors are firm-cluster bootstrap standard deviations of the data moments ($B = 500$).

Goodness of fit. Table III reports the model's fit alongside bootstrapped standard errors of the data moments. The model fits closely on the moments most directly tied to the cost-spreading channel: the Pat/R&D slope is −0.55 in the model versus −0.56 in the data, and the R&D/Rev slope is −0.52 versus −0.61. Aggregate growth, the entry rate, the entry R&D share, and the entrant employment share are all within 16% of their data counterparts.

The largest deviation is on the variance of log employment, where the model produces 0.91 against 1.27 in the data. The gap reflects a limitation that is common to all single-step quality-ladder models. The structure of the model, in which firms grow one line at a time through successful innovation and shrink one line at a time through creative destruction, implies a right tail for the distribution of firm size that is often thinner than the Pareto-like distribution in the data.[18]

[18] Luttmer (2011) shows that matching the Pareto-like right tail requires ingredients beyond the basic Klette–Kortum dynamics used here, such as idiosyncratic productivity shocks or heterogeneous entrant types.



(a) Share of R&D labor devoted to process efficiency

(b) Probability that an attempt against the firm fails

Figure 1: Process efficiency and blocking by firm size

Notes: Both panels are computed at the parameter estimates in Table II, with all subsidy rates set to zero. Firm employment is shown on a log scale. Dashed lines mark the 50th, 90th, and 99th percentiles of the employment distribution across firms.

The calibrated economy. Figure 1 plots two features of the calibrated economy against firm employment: the share of R&D labor that a firm devotes to process efficiency, and the probability that an attempt to take one of its lines fails. Both features depend strongly on size. In Panel (a), the smallest firms, with about three employees, devote 13% of their R&D labor to process efficiency. The share rises to 39% at the median firm, with about 60 employees, to 61% at the 90th percentile of employment, with about 580, and to 70% at the 99th percentile, with about 1,600. At the estimated parameters, the composition of R&D is therefore a first-order margin along which firms differ. In Panel (b), attempts against the smallest firms almost always succeed, whereas about 12.5% of attempts against firms at the 99th percentile fail. Most of this increase takes place between the median and the 90th percentile. This shows how investment in process efficiency by large firms imposes a negative externality on smaller firms, whose innovations against them are more often blocked.

Nontargeted moments. The estimation targets only the average slopes of log(R&D/Rev) and log(Pat/R&D) on log employment, but the shape of these ratios across the size distribution remains non-targeted. Table IV therefore reports these ratios on a quintile by quintile basis, comparing them against their data counterparts. It also does so for log(Pat/Emp) and log(R&D/Emp) in each employment quintile. The comparison is conducted by expressing these ratios as deviations from the smallest quintile [19].

The model reproduces the negative size gradient of all four moments, with the closest fit on the two

[19] Model units (efficiency labor) and data units (nominal yuan, headcount) differ by orders of magnitude, so I report z-score normalized deviations from the smallest quintile to place both sets of moments on a common scale.

Table IV: Nontargeted moments

	R&D/Rev	Pat/R&D	Pat/Emp	R&D/Emp
Panel A: Model				
Quintile 2	−1.55	−0.66	−1.05	−1.51
Quintile 3	−2.14	−1.27	−1.75	−2.11
Quintile 4	−2.55	−1.84	−3.05	−2.54
Quintile 5 (largest)	−2.81	−2.75	−1.77	−2.82
Panel B: Data (Chinese Tax Survey)				
Quintile 2	−2.34	−0.83	−1.29	−1.75
Quintile 3	−2.40	−1.29	−1.99	−2.20
Quintile 4	−2.53	−1.62	−2.32	−2.44
Quintile 5 (largest)	−2.67	−1.96	−2.91	−2.90

Notes: Entries are z-score normalized deviations of each ratio from its value in the smallest quintile, computed separately for model and data. Quintiles are defined by firm counts; the data cutoffs are 14, 58, 133, and 332 employees.

R&D-intensity columns, especially R&D/Emp. The one clear miss is Pat/Emp at the largest quintile, where the model's profile turns back up while the data continue to decline. This can be explained by the model's thin right tail, which generates too little employment dispersion among the largest firms. The top quintile therefore contains firms that are too small relative to their data counterparts, and patents per worker there are correspondingly too high.

4.3 Restricted Model

The negative size gradient of log(Pat/R&D) is a salient feature of the Chinese Tax Survey data. I now show that the model cannot reproduce this gradient without the process-efficiency mechanism.

Write the firm-level patent–R&D ratio as

$$\frac{\mathrm{Pat}(p)}{\mathrm{R\&D}(p)} = \underbrace{\frac{X^*(p)}{h_X(p)}}_{\text{patents per unit of product R\&D}} \times \underbrace{\frac{h_X(p)}{h_X(p)+h_\psi(p)}}_{\text{product share of total R\&D}} . \tag{25}$$

Setting $c_\psi \to \infty$ removes both sources of size dependence. Since process R&D labor $h_\psi(p)$ falls to zero, the second term equals one for every portfolio. Meanwhile, since flow profit reduces to $\pi(Q) = \tilde{p}\,Q$ and no firm invests in process efficiency, every firm faces the same success probability and displacement rate, and the value of an additional line no longer depends on the firm that acquires it. The expected gain $G(p)$ is then the same for every incumbent, so (15) gives the same $X^*(p)$ for every portfolio, and the first term is invariant to size. Therefore, the ratio of patents to R&D does not vary with Q. We cannot recover the empirical slope of −0.56 in the model when process efficiency is shut off.[20]

Process efficiency restores the negative gradient through both factors. Superlinear profit raises $X^*(p)$

[20]The subsidy schedule can introduce size-dependent wedges that generate a small mechanical slope. These are negligible relative to the empirical slope, and the argument abstracts from them.

with Q, so patents per unit of product R&D fall, and under Assumption 2 the product share of R&D falls as well. As a check, I re-estimate the model by setting $c_\psi \to \infty$ and re-adjusting all other parameters. The resulting slope is +0.12, with a sign opposite to the data (Online Appendix Table H.1).

Online Appendix H.3 reports a further validation exercise. It treats the calibrated model as the data-generating process and regresses small-firm patenting on subsidy concentration across nineteen simulated subsidy schedules. The small-firm coefficient is −0.38, slightly over half of the column-3 estimate in Table I.

5 Counterfactual Policy Analysis

China is an empirically relevant case for the blocking externality. Existing subsidies there tend to favor large incumbents, while patent output is concentrated at smaller firms. Online Appendix A documents the two distributions across employment-size bins in the Chinese Tax Survey. Firms with more than 10,000 employees receive 59 percent of innovation subsidies but produce only 12 percent of patents, and patent output per subsidy yuan peaks among mid-sized firms before falling toward zero at the largest. The mismatch in the data between where subsidy flows and where innovation occurs raises two questions: what would happen to innovation and welfare if subsidies were instead allocated differently along the size distribution, and how much of the effect is accounted for by the model's blocking externality?

I answer these questions with three counterfactual exercises using the calibrated model. The first asks whether the current subsidy regime is welfare-improving relative to a counterfactual with no subsidies. The second holds the existing subsidy budget fixed and asks how welfare would change if the planner can freely reallocate it across firms of different sizes. The third allows the planner to determine *both* the size of the subsidy budget and its allocation across firm sizes.

5.1 Setup

The policy instrument is a size-contingent subsidy on the cost of R&D labor. It is the feasible second-best policy for planners because the composition of R&D inside the firm is typically non-contractible, and firm size predicts it. To keep the search space tractable, I partition firms into five groups: those with below-median employment (4 product lines or fewer), employment between the median and 90th percentile (5–8 product lines), employment between the 90th and 99th percentile (9–11 product lines), and employment above the 99th percentile (12 product lines or more), and entrants (those without a successful product innovation). The planner sets a separate subsidy rate for each of these groups:

$$\boldsymbol{\sigma} \;=\; \big(\sigma_{<p50},\; \sigma_{p50\text{–}p90},\; \sigma_{p90\text{–}p99},\; \sigma_{>p99},\; \sigma_e\big). \tag{26}$$

A firm in group k faces an effective skilled wage $(1-\sigma_k)\,w_s$ on both product and process R&D. The subsidy budget is financed by a lump-sum tax T on the representative household, and consumption net of the

financing cost is $C = \hat{Y} - T$.[21]

Welfare is evaluated on the balanced growth path. I report the consumption-equivalent variation (CEV), the proportional change in baseline consumption that would make the household indifferent between the baseline and the alternative policy, in percent. Online Appendix H.4 gives the formula.

I construct the baseline subsidy schedule from the Chinese Tax Survey by aggregating subsidies and R&D spending across firm-years within each percentile bin and computing the ratio. The result, in percent, is $\boldsymbol{\sigma}^{\text{baseline}} = (10.0, 7.80, 6.90, 26.1, 0)$. This means the smallest incumbents receive a 10% discount on R&D labor, the top 1% firms receive a 26% discount, and entrants are unsubsidized. Existing subsidies are heavily tilted toward the largest firms.

5.2 Policy experiments

Table V reports the baseline equilibrium alongside three counterfactual policies: removing all subsidies, the welfare-maximizing allocation subject to the baseline budget, and the unconstrained optimum.

Removing all subsidies. Setting $\boldsymbol{\sigma} = 0$ raises welfare by 1.02% and growth by 0.01 percentage points. The current subsidy regime is therefore net negative for both growth and welfare. This is because removing subsidies releases skilled-labor demand, and the skilled wage falls from 0.316 to 0.287. Because small incumbents and entrants spend a larger share of their R&D budget on product innovation than large incumbents do, the lower wage produces more product innovation at small firms than it forgoes at large ones. This increases the aggregate attempt rate $\bar{\tau}$ from 0.0227 to 0.0229.

Budget-neutral reallocation. The subsidy schedule here is $\boldsymbol{\sigma}^{\text{BN}} = (0, 0, 0, 0, 74.8)$. This means the planner withdraws all subsidies from incumbents and passes them to entrants. Welfare rises by 10.7% while growth rises by 0.50 percentage points. The subsidy rate for entrants is high because they spend their entire R&D budget on product innovation. Process efficiency is unattractive to them since they have no existing lines to spread process improvements over. Under the budget-neutral reallocation, the entrant success probability rises from 0.54 to 0.64 because the lines entrants draw are held by smaller, less process-efficient incumbents on average.

Variable-budget reallocation. With the budget constraint removed, the optimal policy involves spending 5.72% of GDP on R&D subsidies, seven times over the baseline budget of 0.761%. The subsidy schedule here is $\boldsymbol{\sigma}^{\text{VB}} = (28.5, 0, 0, 0, 91.4)$, meaning that entrants and the smallest incumbents are subsidized at 91.4% and 28.5% respectively. Medium and large firms receive nothing. This is because the marginal dollar of subsidy spent on them goes mostly to process efficiency, which strengthens the blocking externality and slows creative destruction. Under this counterfactual, small firms generate 81% of all patents, while large

[21] Partitioning firms into four groups rather than allowing a separate rate for each portfolio also gives the reallocation gains a lower-bound interpretation, since a finer partition would give the planner more flexibility.

Table V: Counterfactual policy experiments

	Baseline	No Subsidy	Budget-Neutral Optimum	Variable-Budget Optimum
Panel A: Subsidy schedule and fiscal cost (%)				
$\sigma_{<p50}$	10.0	0	0	28.5
$\sigma_{p50\text{–}p90}$	7.80	0	0	0
$\sigma_{p90\text{–}p99}$	6.90	0	0	0
$\sigma_{>p99}$	26.1	0	0	0
σ_{entrant}	0	0	74.8	91.4
T/Y	0.761	0	0.766	5.72
Panel B: Margins of innovation				
$\bar{\tau}$	0.0227	0.0229	0.0277	0.0333
B_e	0.541	0.536	0.643	0.819
w_s	0.316	0.287	0.283	0.315
$\bar{n}$	3.95	3.93	3.05	2.35
Panel C: Patent shares by firm size (%)				
Small ($n < 5$)	37.7	38.0	56.4	81.0
Medium ($5 \leq n < 12$)	57.7	60.0	42.2	15.8
Large ($n \geq 12$)	4.28	1.61	0.30	0.01
Panel D: Growth and welfare				
g (%)	2.31	2.32	2.81	3.47
Δg vs. baseline (pp)	0	0.01	0.50	1.16
CEV $\xi - 1$ (%)	0	1.02	10.7	18.2

Notes: Panel A is in percent, with T/Y the subsidy budget as a share of GDP; all entries are to three significant figures. $\bar{\tau}$ is the aggregate attempt rate per product line, B_e the entrant success probability, w_s the skilled wage, and $\bar{n}$ the average number of lines per firm. Panel C shares are successful product innovations by size bin as a share of the total; the entrant share (0.4% at baseline, 3.2% under the variable-budget optimum) is omitted. Percentile bins are fixed at the no-subsidy distribution. CEV is the consumption-equivalent welfare change relative to baseline, with consumption net of the fiscal cost.

firms generate nearly zero. The aggregate attempt rate $\bar{\tau}$ rises from 0.0227 to 0.0333. Welfare rises by 18.2% and growth by 1.16 percentage points.

5.3 Growth decomposition and the welfare cost of blocking

The welfare numbers in Table V are the net effect of three forces, which I separate by decomposing realized growth as

$$g = \underbrace{\lambda X_e}_{\text{gross entry attempts}} + \underbrace{\lambda \int X^*(p)\, dM(p)}_{\text{gross incumbent attempts}} - \underbrace{\lambda \Phi_A \bar{\tau} (1 - \bar{B})}_{\text{blocking loss}}. \tag{27}$$

The first two terms are the gross rates at which entry and incumbent innovation attempts arrive, each scaled by the quality step λ. The third term is the rate at which attempts are blocked: Φ_A is the share of active lines, $\bar{\tau}$ is the aggregate attempt rate, and $1 - \bar{B}$ is the average probability that an attempt on an active line fails. Table VI reports each component across the baseline and the three counterfactuals.

Table VI: Growth decomposition and the welfare cost of blocking

	Baseline	No Subsidy	Budget-Neutral Reallocation	Variable-Budget Reallocation
Panel A: Decomposition of realized growth (%)				
Gross entry attempts (λX_e)	0.305	0.313	0.491	0.702
Gross incumbent attempts ($\lambda \int X^* dM$)	2.146	2.154	2.493	2.894
− Blocking loss ($\lambda \Phi_A \bar{\tau} (1 - \bar{B})$)	0.141	0.146	0.176	0.128
Realized growth g	2.310	2.322	2.808	3.469
Panel B: Welfare cost of the blocking externality				
Innovation attempts blocked (%)	5.74	5.90	5.88	3.55
Permanent CEV cost of blocking (%)	3.27	3.36	3.66	2.32

Notes: Panel A reports the components of equation (27) in percentage points. Panel B reports the share of innovation attempts blocked and the welfare cost of blocking under each policy.

Panel A shows that the gains from the counterfactuals come mainly from gross innovation attempts, and only in the variable-budget rellocation it also comes from a smaller blocking loss. Under the budget-neutral and variable-budget optima, entry attempts rise directly with the entry subsidies, and incumbent attempts rise through the wage channel: subsidies withdrawn from the top release skilled labor to the smaller firms that were previously crowded out.

Panel B reports that under the benchmark regime, blocking destroys 5.74% of all innovation attempts and imposes a welfare cost of 3.27%.[22] Meanwhile, how the blocking loss moves across the three counterfactuals depends on two opposing forces: removing subsidies from large incumbents lowers their process efficiency and hence the blocking probability on the lines they hold, while the lower skilled wage lets medium-sized firms, which received little subsidy at baseline, invest more in process efficiency on their own.

Which force dominates is a quantitative question. Under the no-subsidy and budget-neutral counterfactuals, the second force dominates. In the budget-neutral counterfactual specifically, the high entry subsidy generates more attempts from entrants, which leads to more of them being blocked and hence a greater blocking loss. The variable-budget optimum is the only policy under which the blocking loss falls. Its heavy entry and small-incumbent subsidies compress the firm-size distribution, shrinking the size of the average firm from 3.95 to 2.35 product lines. With fewer product lines, incumbents invest less in process efficiency, so blocking loss falls.

5.4 Robustness

The conclusions are stable across three robustness exercises: two that bring additional moments into the estimation, and one that varies the externally calibrated curvature of the R&D production function. Online Appendix J reports the parameter estimates and counterfactual outcomes for each.

[22] The model generates a thinner right tail than the empirical firm-size distribution (footnote 18). Because the model assigns the largest blocking effect to the top of the size distribution, a better right-tail fit would likely raise the blocking welfare cost. The 3.27% figure is therefore a lower bound.

The first exercise adds the size gradient of patent output per subsidy yuan, documented in Online Appendix A but nontargeted in the benchmark estimation, as an eighth moment. The estimates move modestly ($\hat{\omega} = 3.20$ versus 3.28 in the benchmark), and the three counterfactual welfare gains are 1.1%, 12.3%, and 21.5%.

The second exercise targets the process-efficiency channel directly rather than through the Pat/R&D gradient, adding as an eighth moment the share of large product innovators that also conduct process innovation in the National Enterprise Innovation Survey (87.1%, Section 3.3). The benchmark excludes this moment because it measures an extensive margin, while the model's R&D composition choice is an intensive margin. The augmented estimates are within one standard error of the benchmark, and the welfare gains are within 0.2 percentage points.

The third exercise re-estimates the model at $\gamma = 0.4$ and $\gamma = 0.6$, since γ governs how strongly innovation responds to R&D input and the welfare gains are likely to be sensitive to it. Parameter estimates remain within roughly one standard error of the benchmark. The welfare gains are 1.0–1.1%, 12.6–13.6%, and 21.4–22.4% across the two values, and the optimal schedules keep their shape, with the budget concentrated on entrants and the smallest incumbents.

Moreover, I extend the model with firm-type heterogeneity as in Acemoglu et al. (2018), but with permanent types: each firm draws a type $\theta_f \in \{\theta^h, \theta^l\}$ upon successful entry. Process efficiency, the blocking condition, the displacement rate, and the success probability remain functions of a firm's aggregate quality alone, so the composition gradient and the blocking externality apply to both types at any given size (Online Appendix K). A planner who conditions only on size cannot exploit type heterogeneity, so the welfare gains reported above measure the size-margin component of the total reallocation gain. Re-estimating the model with types is left for a future revision.

6 Conclusion

This paper develops a quality-ladder growth model in which product quality and process efficiency jointly determine a firm's competitiveness on each product line. Because the returns to process efficiency scale with firm size while its cost does not, large firms tilt their R&D toward process improvement, and the resulting blocking externality makes their lines harder to displace. Subsidies to large firms strengthen this advantage, and in the Chinese data, small-firm patenting is lower in industries where a larger share of subsidies goes to large firms. Calibrated to Chinese data, the model implies that the current subsidy regime is welfare-reducing relative to a regime with no subsidies. Reallocating the existing budget away from large incumbents toward entrants raises welfare by 10.7%, and a variable-budget allocation reaches 18.2%.

The mechanism is not specific to China. Cost-spreading is a property of the technology of process improvement, and in any economy where management practices, supply-chain investments, and organizational capital apply uniformly across a firm's product lines, larger firms will tilt R&D toward process efficiency. The blocking externality follows in any quality-ladder environment where entrants must surpass incumbents

on both quality and cost. This yields the policy implication that innovation subsidies favor small firms and entrants not because they are more capable but because the composition of their R&D is closer to what drives growth.

I leave two questions for future work. First, the welfare gains reported here come from the firm-size margin alone. A planner who could also condition on founder ability, industry, or product characteristics would likely achieve larger gains from an optimal policy. Second, this paper infers the composition of R&D from the size gradient of patents per unit of R&D. Data that measure it directly, such as innovation surveys that separate process from product spending, or process patents identified from their claims, would allow the composition margin to be estimated rather than inferred.

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